

Impact Evidence Pack

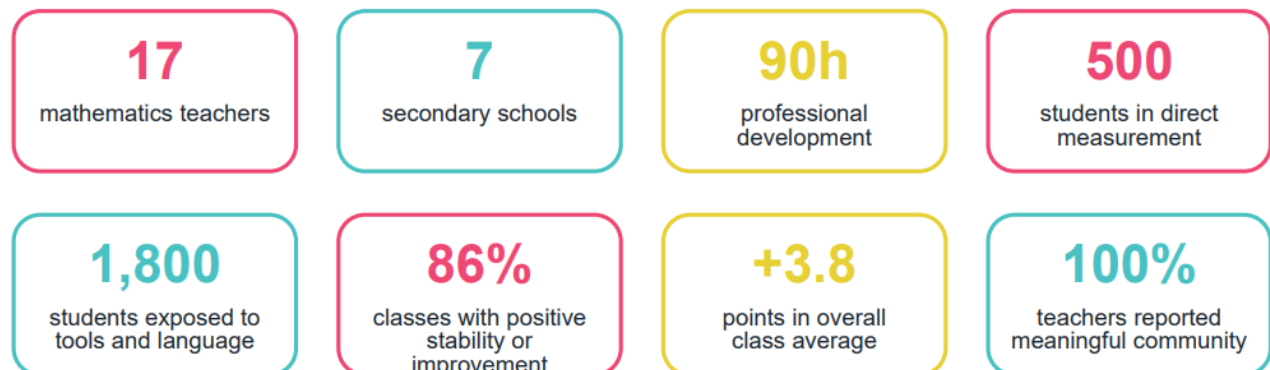
Empowering Mathematical Mindset: A Citywide Teacher Leadership Model

Purpose:

This evidence pack supports the Impact / Potential Impact section of the QS Reimagine Education Awards application. It summarises how the initiative addresses the educational challenge, the number and quality of participants engaged, and the learning, behavioural and institutional outcomes observed in the pilot.

Problem:	Teacher Leadership:	Classroom Routines:	Student Behaviours:	Impact:
Math anxiety, low confidence, avoidance	90-hour training, mentoring, observations	mistakes, self-assessment, goals	readiness, help-seeking, responsibility	positive stability, achievement growth, citywide scale

Impact mechanism: changing daily learning behaviours before, during and after mathematical challenge. The project measures not only scores, but also the learning conditions that make progress possible.



Core impact claim. The project creates impact by strengthening the learning behaviours that precede achievement: persistence, readiness, help-seeking, reflection, responsibility and the ability to learn from mistakes.

1. How the project addresses the problem

Many students experience mathematics as a space of fear, failure and low self-confidence. In such conditions, students may avoid challenge, stop asking questions, disengage from practice and interpret mistakes as evidence that they are not “good at mathematics”. The initiative responds by changing the daily learning culture of mathematics classrooms.

Teachers are trained to embed mindset-based routines into ordinary instruction: learning from mistakes, self-assessment, goal setting, short success checks, reflection, differentiated tasks and language that links effort, strategy and progress. These tools make the mental and emotional conditions of learning visible and teachable.

The approach is effective because it works through teachers, the most stable agents of change in the school system. Rather than delivering a short student workshop, the model builds teacher leadership, classroom implementation, mentoring, observation and data-informed refinement. As a result, students encounter the mindset language repeatedly and in direct connection with mathematical tasks, tests, errors and improvement.

Why mathematics requires a mindset-based approach: mathematics is a strategic gateway to advanced study, STEM pathways and future opportunities, yet it is also one of the subjects most strongly associated with anxiety, avoidance and fixed ability beliefs. In the Be'er Sheva context, the advanced five-unit mathematics matriculation rate has remained around 15% in recent years, and many teachers identify mathematics anxiety and negative self-perception as key barriers preventing students from entering or persisting in advanced mathematics. The project therefore treats achievement not only as a content issue, but as a learning culture issue: students need confidence, resilience, self-regulation and the ability to learn from mistakes in order to stay engaged with demanding mathematical learning.

Why the citywide level matters. Single-school initiatives may improve local practice but rarely change the professional culture across a system. This model builds a municipal network of trained teachers who use aligned tools and a shared language, allowing impact to grow from one classroom to multiple classrooms, from one school to several schools and from a pilot cohort to a sustainable urban learning infrastructure.

2. Engagement to date and quality of engagement

The pilot engaged 17 mathematics teachers from 7 secondary schools in Be'er Sheva. Teachers participated in a 90-hour professional development and mentoring process that included training meetings, classroom implementation, individual mentoring, observations, reflection and evidence collection.

17 teachers trained as mindset-based learning leaders	7 secondary schools participated	90h training, mentoring and classroom implementation	500 students included in direct measurement
1,800 students exposed through participating teachers	100% teachers reported sharing, openness and learning	94% rated facilitation as professional and clear	88% reported practical classroom tools

The quality of engagement was high because participation moved beyond attendance. Teachers implemented tools in their classrooms, selected a class for systematic measurement, used reflection and assessment instruments, documented student responses and discussed implementation challenges. The project therefore created a professional learning community rather than a one-time training event.

Teachers reported that the process changed how they viewed their role as mathematics teachers: not only as transmitters of content, but as leaders of confidence, persistence, reflection and responsibility in learning.

Evidence type	Quote
Teacher evidence	"This was not another professional development course. It changed how I think about my role as a mathematics teacher."
Classroom evidence	"I started using mistakes not only as something to correct, but as a starting point for learning."
Student engagement evidence	"Students who usually avoid participation became more engaged when we began speaking about effort, progress and mistakes."

3. Learning outcomes and evidence of positive impact

The project contributes to positive learning by developing students' resilience, self-efficacy, responsibility, flexible thinking and ability to learn from mistakes. These are not treated as general values, but as daily learning behaviours embedded in mathematics instruction.

Teacher-reported student behaviour changes

Student behaviour change	Result
Readier for assessments	82%
More willing to address errors	76%
More help-seeking	71%
More responsibility for learning	71%
More persistence when challenged	69%
More self-confidence in math	65%

Percentages indicate teachers reporting medium or higher positive change among students.

Teacher reports indicated that students became readier for assessments, more willing to examine errors, more likely to ask for help and more responsible for their learning. These behaviours are important because they describe the student actions that enable improvement when mathematics becomes difficult.

The pilot also showed positive achievement indicators: 86% of measured classes demonstrated improvement or positive stability, and the overall class average increased from 67.3 to 71.1.

These findings are presented as pilot indicators and not as causal proof, since the first year did not include a control group or common test. They nevertheless support the need for continued, more systematic measurement.

Achievement indicator	Result
Overall class average	Increased from 67.3 to 71.1, a +3.8 point change.
Measured classes	86% showed improvement or positive stability.

4. Impact assessment and future measurement

The evaluation model combines quantitative and qualitative evidence. It examines not only whether grades change, but also how learning behaviour, classroom language, teacher practice and student responsibility change over time.

Impact dimension	Evidence collected	Future measurement
Student learning behaviours	Readiness, help-seeking, responsibility, persistence, error analysis	Student questionnaires, self-assessment tools, classroom evidence
Academic progress	Class averages, movement across achievement thresholds	Common achievement indicators, repeated cycles, comparison groups
Teacher practice	Implementation of routines, reflection, practical tools	Observation, teacher surveys, implementation portfolios
Institutional outcomes	Shared professional language, school-based leadership, municipal network	School implementation files, citywide reporting, scaling indicators

Projected impact. In the next stage, the model is designed to create a multiplier effect. The first cohort of trained teachers continues as school-based leaders, a new teacher cohort enters training, and each participating school builds a local implementation plan. This structure allows the initiative to reach additional teachers and students while strengthening the municipal infrastructure for mathematics learning.

Professional conclusion. The pilot indicates that Empowering Mathematical Mindset can create meaningful impact at three levels: student learning behaviours, teacher professional identity and citywide institutional capacity. The next phase will strengthen the evidence base through more consistent measurement, comparison across cycles and clearer tracking of student-level progress.